

IT Applications in Business

Blockchain and cryptocurrencies

Marek Woda, PhD



HR EXCELLENCE IN RESEARCH



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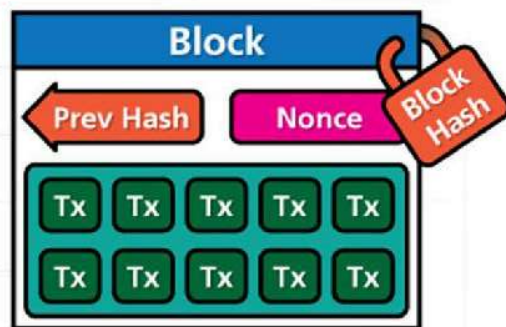
Lecture 3: ***Blockchain and Cryptocurrencies***

1. **BTC scaling issue**
2. **ICO & DAO**
3. **Consensus mechanisms**
4. **Smart contracts**
5. **Blockchain applications**

Bitcoin – scaling issue

- Each block ~2k transactions => 1 MB in size
- A new block every 10 minutes (pre-set interval) regulated by mining difficulty

$$\frac{2000/10}{60} = 3,2 \text{ TPS}$$



Tx = 200 transactions

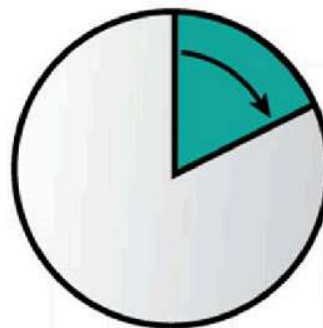
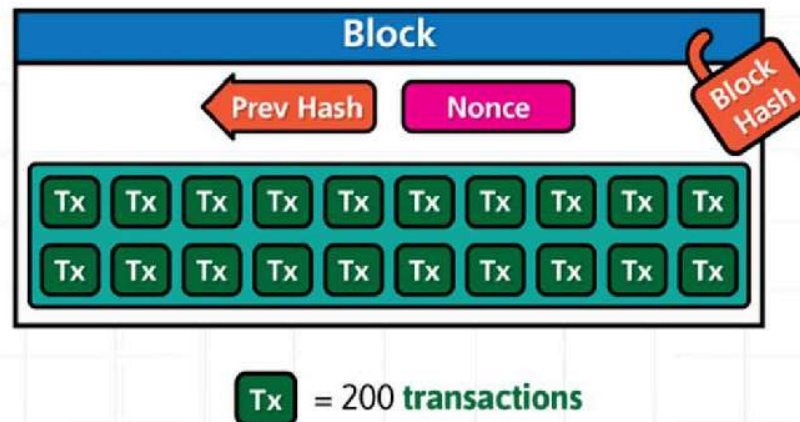


Illustration by CryptoGraphics.Info

- Note: VISA does around on average 2k /second

Bitcoin – scaling issue – possible solutions

- **Doubling the block size (“2x”)**
 - Gain: 2x speed = 6,4 TPS
 - would make the blockchain **grow exponentially larger**, too large to handle in a few years time



Bitcoin – scaling issue – possible solutions

- Segregated witness (“SegWit”)

- Gain: 2.6x speed = 8,3 TPS

- each block contains **transactions** and **signatures** (hashes)

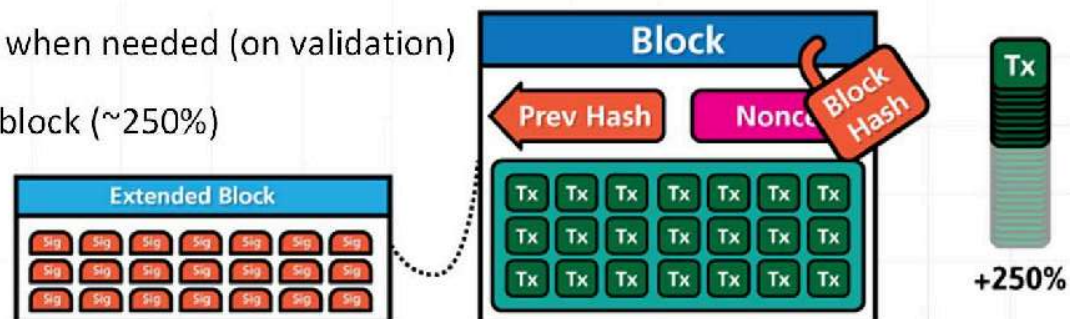
- signatures (roughly 60% of a block’s size) are only needed for verification,

- full blockchain copy carries around 60% ‘**dead weight**’

- separation (‘segregating’) of signatures (‘witnesses’) in an optional ‘extended block’

- extended block will then be requested when needed (on validation)

- more room for transactions inside the block (~250%)



Tx = 200 transactions



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Bitcoin – scaling issue – possible solutions

- *Schnorr signatures*

- Gain: 2x speed, 5.2x when combined with SegWit
- mathematical rules that link → **private key / public key / signature**
- many transactions include multiple inputs, referring to the address's bitcoins are sent from
- each require the signature of the sender, and all these signatures are included in the transaction

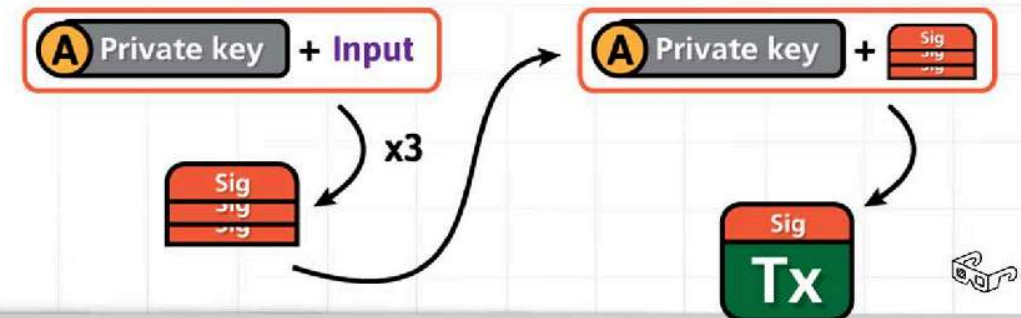
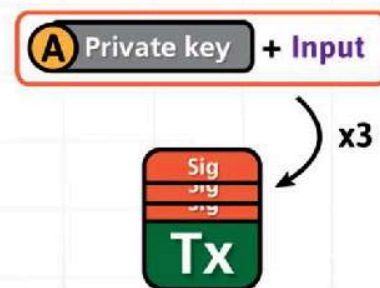
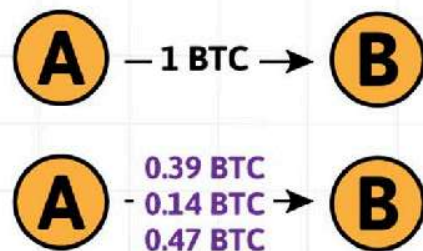


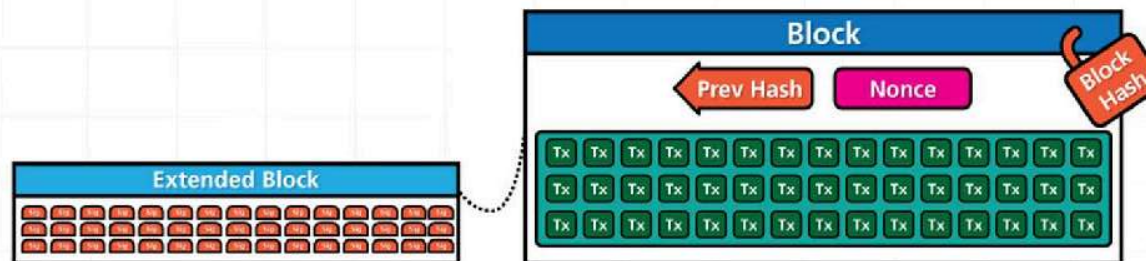
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Bitcoin – scaling issue – possible solutions

- Segregated witness + double block size: “SegWit2x”
 - Gain: 5.2x speed = 16,6 TPS
 - a hybrid = double block size with Segregated Witness

$$\frac{2000 * 5/10}{60} = 16,6 \text{ TPS}$$

- SegWit2x suffers from the same **growth-related problem** as doubling the block size



Tx = 200 transactions

Bitcoin – scaling issue – possible solutions

- **Lightning Network**
 - unaltered blockchain (same block size, *PoW* → mining, etc.)
 - instead of every transaction being stored - **payment channels** are used
 - direct *p2p* connections between nodes (users) on the network

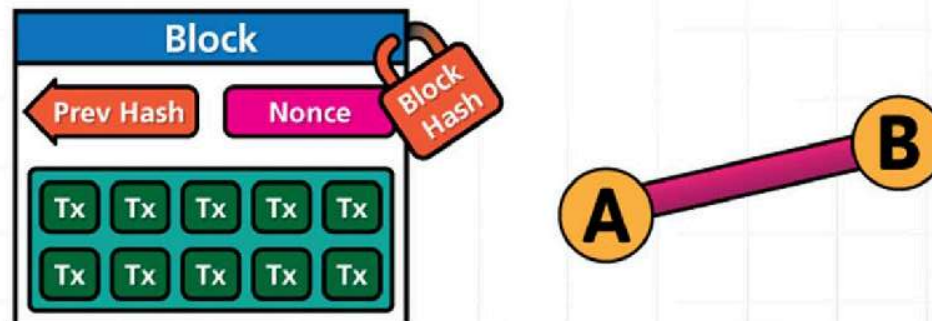
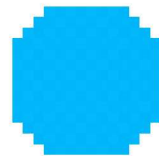


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Bitcoin – scaling issue – possible solutions

- **Lightning Network**

- resulting transaction is handled **on-chain** by the network as usual
- if a transaction in the channel is malformed or invalid, all transactions made in the channel → will fail
- **ALL** nodes can freely create channels
- no time-limit for channels, so these can stay open **indefinitely** – or until funds are to be released, upon which (one of) the nodes close(s) the channel

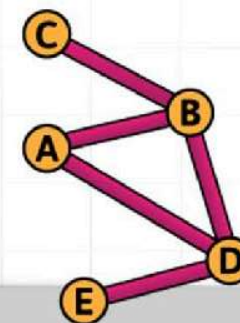
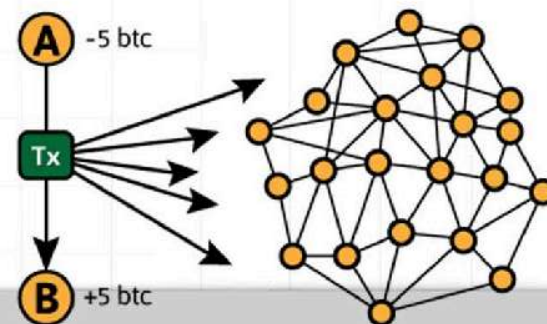


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Bitcoin – scaling issue – possible solutions

- **Lightning Network**

- in **theory**, it will be able to handle *millions to billions* of transactions per second (TPS)
- this blows away legacy systems, such as VISA, by many orders of magnitude

BTC 3-16 TPS

Ethereum 20 TPS

PayPal 200 TPS

SWIFT ~200-350 TPS

Visa 24 000-56000 TPS

SegWit



VISA

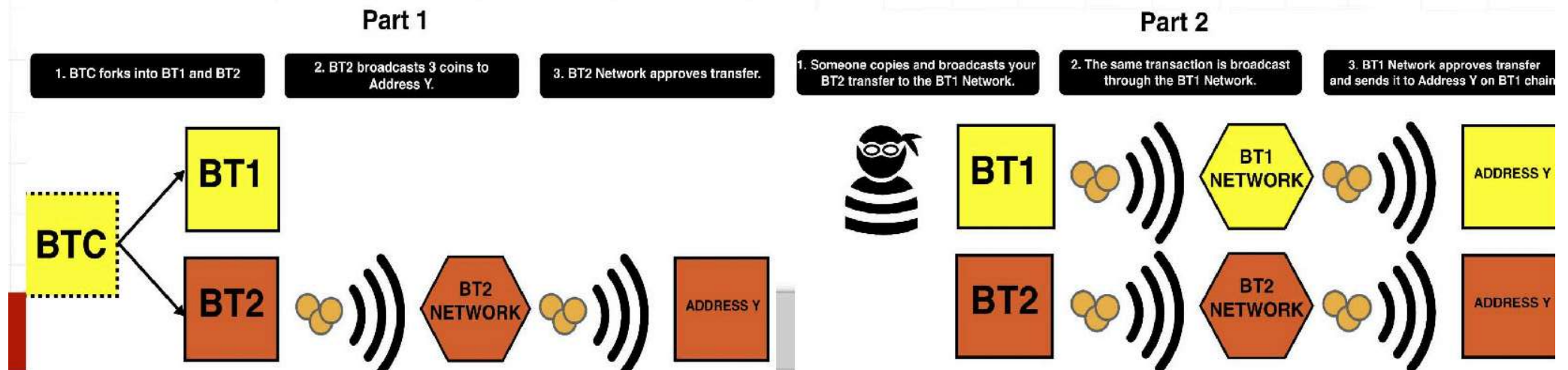
LN



Replay attack = playback attack

- **Double-spending**

- an exploit that can occur when two ***forked*** crypto-currencies allow transactions to be valid across both chains
- a potential flaw in a digital cash scheme in which the same single token can be spent more than once
- possible because a digital token consists of a digital file that can be duplicated or falsified

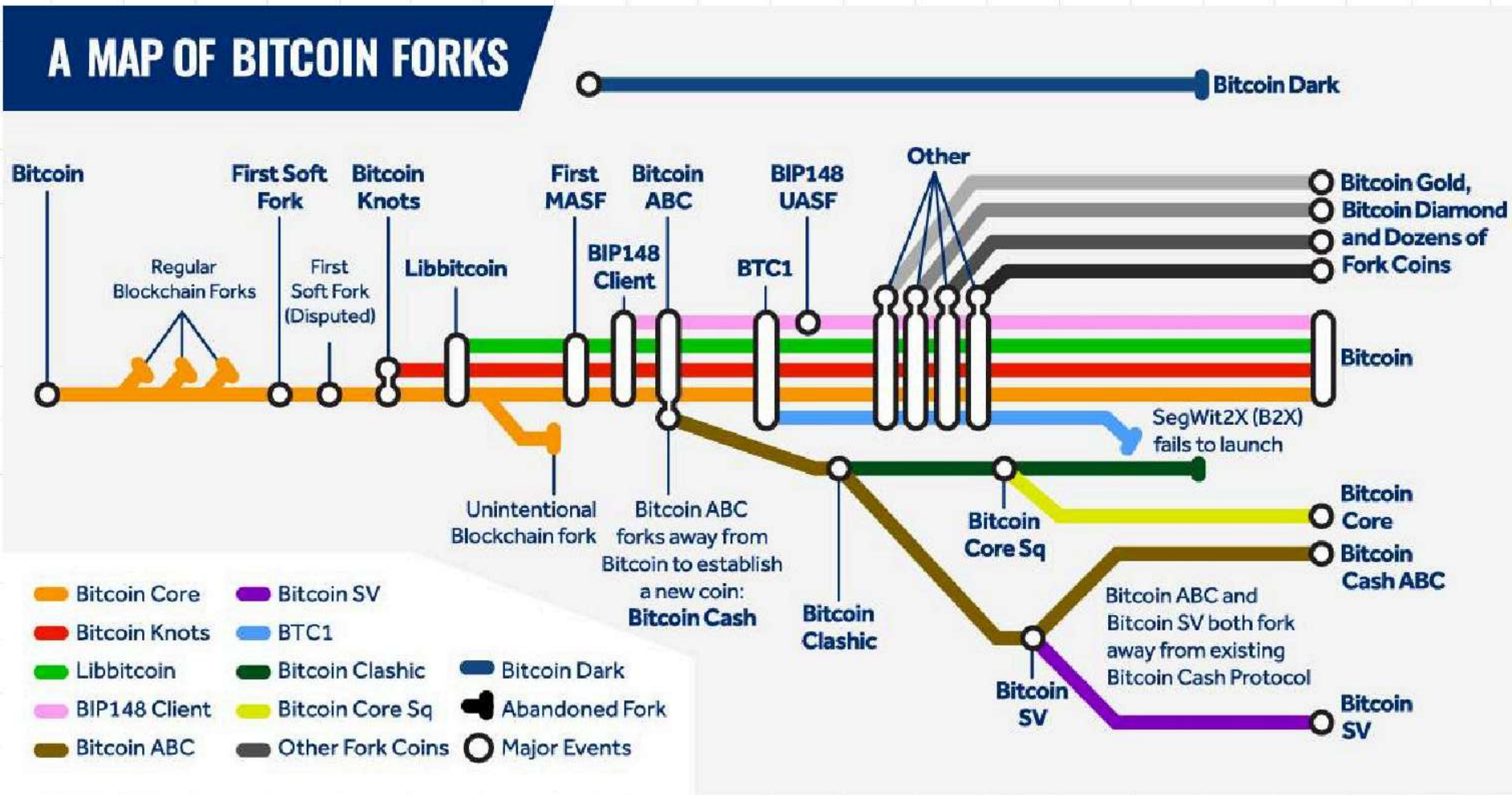


Hard Fork

- Not backward compatible
- Old nodes don't accept new blocks
- Nodes that run old versions will see new transactions as invalid
- Controversial
- Requires everyone to upgrade



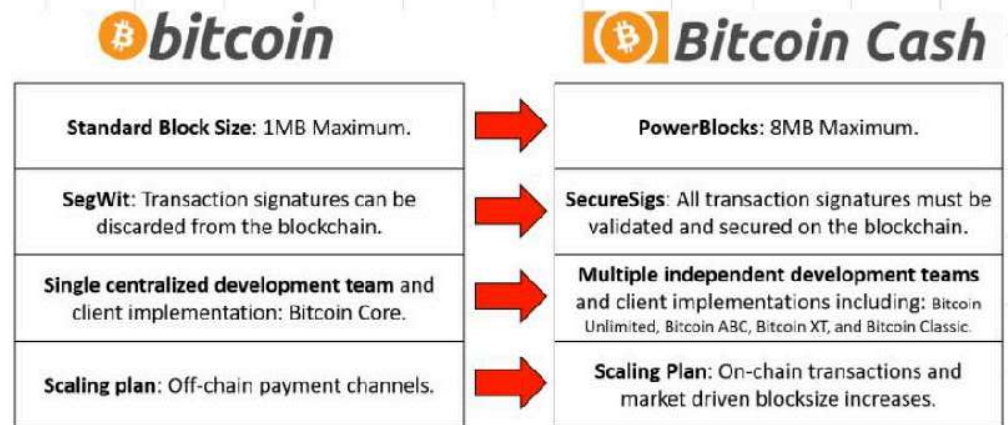
A MAP OF BITCOIN FORKS



<https://www.visualcapitalist.com/major-bitcoin-forks-subway-map/>

BitCoin Cash (BCC / BCH)

- It is NOT a cash equivalent of BTC
- another new cryptocurrency
- 1st August 2017 – BCC was created as *a hard fork* from BTC
 - block size increases to 8MB



Consensus mechanisms

- systems of reaching agreement on the state of the blockchain

- protocols for authenticating new entries on a blockchain / governing changes:

- a. Proof of Work (PoW)**

- operates on the principle that it is expensive to add a block of new transactions to the blockchain
 - *Bitcoin, Litecoin, Ethereum (pre-Casper)*

- b. Proof of Stake (PoS)**

- operates on the principle that 'everyone' can win the chance to add a block to the blockchain, by using a system not much unlike a *lottery*
 - *Ethereum (Casper), Peercoin, NxtCoin, BlackCoin*



Consensus mechanisms

- protocols for authenticating new entries on a blockchain / governing changes

3. Leased Proof-of-Stake (LPoS)

- operates on the principle that literally 'everyone' can win the chance to add a block to the blockchain, by using a system not much unlike a lottery which can be participated in using leased coins
- Waves

4. Delegated Proof of Stake (DPoS)

- operates on the principle that 'everyone' can win the chance to add a block to the blockchain, by using an election followed by a lottery.
- BitShares

5. Delegated Byzantine Fault Tolerance (dBFT)

- operates on the principle of **Byzantine Fault Tolerance** to verify blocks, by using an election followed by a validation process
- NEO

Proof of Work (PoW)

- ***Mining (calculation, hashing process)***

- requires significant computational power to solve complex cryptographic-based equations before they're authorized to add data to a blockchain-based cryptocurrency
- process of calculation hash based on known values inside the block:
 - previous block's hash, hashes of the transactions contained in the block (*Merkle root*), the block's creation time
 - these values are fixed → hashing always produce the same result
- to be able to alter the hash outcome, a 4th (*unknown*) string is added: the so-called '**nonce**'
- continuously randomizing this nonce, the node can find the correct hash that fits the difficulty pattern

Proof of Work (PoW)

- **process in a nutshell**

- collect transactions from the network
- validate them, and doesn't allow conflicting ones
- put them into large bundles called **blocks**
- compute ***cryptographic hashes*** over and over until it finds one "*good enough to count*"
- then submit the block to the network, adding it to the block chain and earning a reward in return

Proof of Work (PoW)

- **Mining** (calculation, hashing process)

- once the right **nonce** is found, it is placed inside the block, and the calculated hash (signature) is added to the block's header, securing it and its contents
- the hash acts as a sort of lock, locking the block, preventing further changes of the transactions stored inside (as changing just one character inside the block would produce a completely different hash)

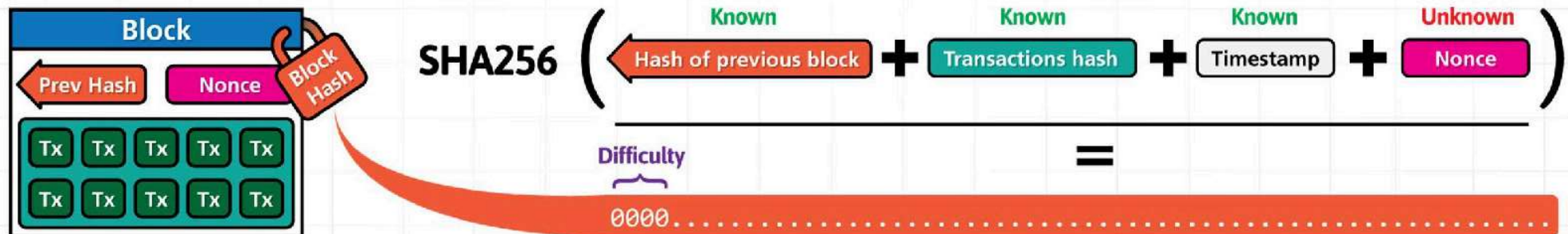


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Proof of Work (PoW)

- process in detail

- miners must compute a **cryptographic hash** of the block that meets certain criteria
- namely to find **nonce** so that

$$\text{sha256}(\text{sha256}(\text{data} + \text{nonce})) = \text{difficulty}$$

- ✓ **nonce** - integer number the miner chooses freely (this choosing of the **nonce** and checking if the condition ($< \text{difficulty}$) is met comprises the work
- ✓ **data** is a **hash** over the contents of the block (transactions) and the previous block's hash
- ✓ **sha256()** - SHA-256 cryptographic hashing function
- ✓ **difficulty** - value adjusted consensually by the nodes of the BTC network to adjust for change in network computation power to have one block every 10 minutes found by the network

Proof of Work (PoW)

- **State of Consensus**

- node that found the *nonce* first, it broadcasts the complete block to all other nodes
- nodes can then verify the work that was done (the nonce can thus be considered as the 'proof of work') this **MUST** then produce the same hash (signature). If it does, block is valid, and added it to the blockchain, nodes discard their own blocks
- If more than **50%** of the nodes do this, a **state of consensus** is reached, and the block is **valid**
- In the course of time, the entire network will have added this block to its blockchain.
- The node that found the nonce will then receive an award (i.e. Bitcoins)

Proof of Work (PoW)

- **Difficulty**

- To compensate for *increasing hardware speed* and varying interest in running nodes over time, the **difficulty** is determined by a moving average targeting an average number of blocks per hour.
- In Bitcoin's case, this makes sure a new block becomes available roughly every **10** minutes
- Changes every 2016 blocks (approx. 2 weeks)
 - If blocks are generated too **fast**, the difficulty increases
 - If they are generated too **slow**, the difficulty decreases

$$Size = [1, 2^{256}]$$

Proof of Stake (PoS)

- *In a nutshell*

- all coins are created in the beginning and never change → *no mining*
- a new block will be added by a **randomly** picked node (much like a *lottery*)
- each of nodes can choose to invest a certain number of coins (**stake**) into a deposit, which is basically like buying lottery tickets
- coins in deposit are not spendable

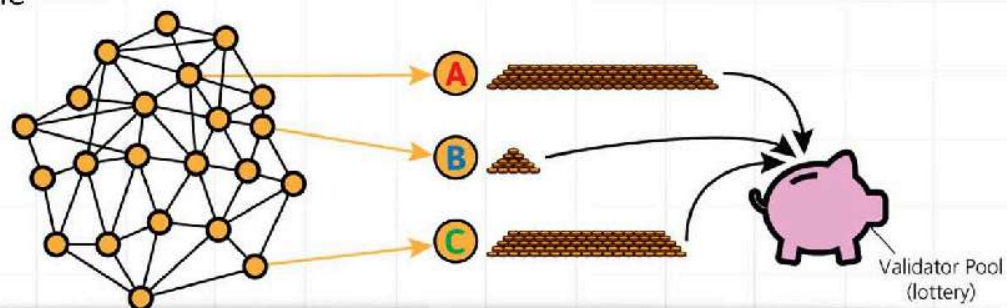


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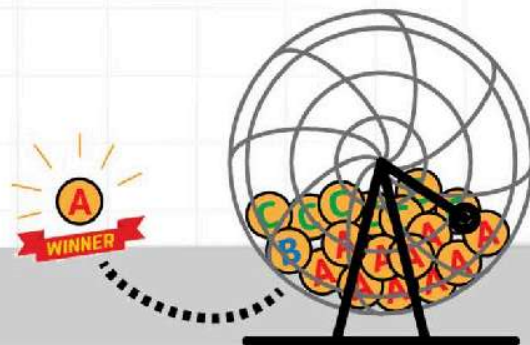


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Proof of Stake (PoS)

- **Consensus**

- more coins a node invests, the higher its stake → the higher the chance of winning the 'lottery'*
- winner is determined by the **consensus algorithm** (a '**lottery machine**')
- **lottery's prize** → rights to collect waiting transactions into a block and collect the transaction fees
- winning node provides the new block with a reference to the preceding block and is then broadcast, upon which all nodes will verify the transactions inside and add the block to their copy of the blockchain
- as there is no computational-intensive mining with Proof-of-Stake - **very energy efficient**



Transaction fees
(existing coins)



Leased Proof of Stake (LPoS)

- *In a nutshell*

- works exactly like *PoS*, but uses leasing to provide small holders with an incentive to take part
- low holding nodes (*D, E, F, G, H*) lease their balance to a staking node (*B*)
- The leased funds remain in full control of the holder

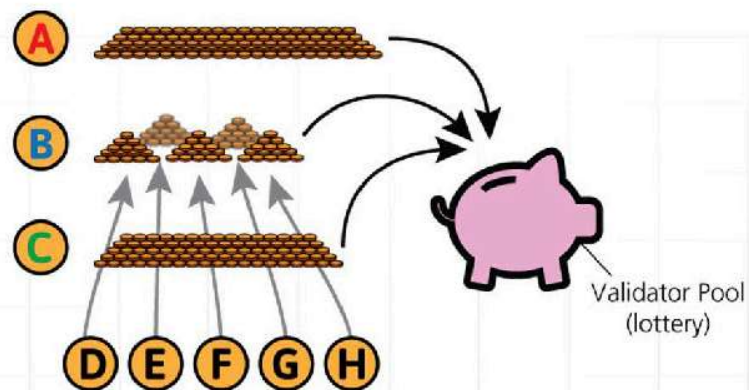


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Leased Proof of Stake (LPoS)

- **Consensus**

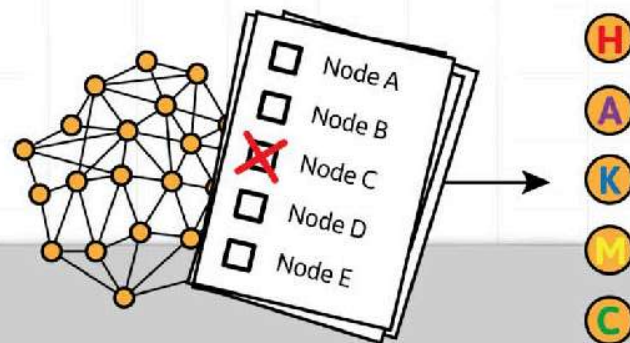
- The leased coins increase the staking node's stake, in turn increasing its chance of winning the 'lottery'.
- Upon winning, the staking node collects waiting transactions into a block and collects the transaction fees, which are shared – **proportionally** – between the staking node and its leasing nodes.
- Operation resumes as in regular **PoS**



Delegated Proof of Stake (LPoS)

- *In a nutshell*

- like LPoS, but coin holders use their balances to elect a list of nodes, regardless of their balance, that will have the opportunity to take part in the 'lottery' → engages **all coin holders**, though it does not reward them directly in the same way as LPoS does
- all nodes in the network, elect a group of nodes to partake in a 'lottery'
- votes from nodes with a **higher balance will have more weight** in the election
- holders can also vote on changes to network parameters, giving them greater influence and ownership over the network



Delegated Proof of Stake (LPoS)

- **Consensus**

- **elected nodes** partake in the 'lottery', on equal chances
- lottery's prize is the rights to collect waiting transactions into a block and collect the transaction fees
- as in regular **PoS**, the winning node provides the new block with a reference to the preceding block and then broadcasts it to all other nodes, **which will then verify** the transactions inside and add the block to their copy of the blockchain
- re-election is held for the next block



IPO vs. ICO

- ***Initial Public Offering (IPO)***

- type of public offering in which shares of a company are sold to investors
- fundraising mechanism
- regulated
- proven and trusted
- later stage



IPO vs. ICO

- **Initial Coin Offering (ICO)**

- crowdfunding centered around cryptocurrency
- a new cryptocurrency/idea is sold in form of "tokens" for legal tender or cryptocurrencies
- tokens supposedly become functional units of currency when the ICO's funding goal is met
- mostly not regulated
- early stages
- a mechanism for scams



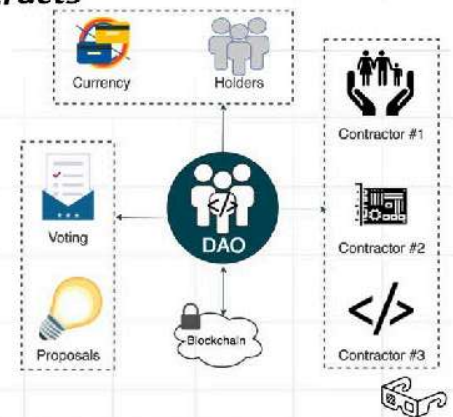
ICO – what to check



1. Read the ICO white paper thoroughly. Does the concept make sense to you?
2. What problem is the product solving? Does it make business sense?
3. Study the team and their experience. Research representatives.
4. Check crypto forums (e.g., subreddit)
5. Look at what rating companies are saying about the ICO

Decentralized Autonomous Organization

- **DAO** known as well as **Decentralized Autonomous Corporation (DAC)**
 - *a transparent, automated, and member-driven organization where rules and decisions are encoded in **smart contracts**, enabling decentralized collaboration and governance without traditional hierarchical management*
 - an organization that is run through rules encoded as computer programs called **smart contracts**
 - a form of investor-directed venture capital fund
 - provide a decentralized business model for organizing enterprises
 - claimed to be completely transparent
 - eliminates the fraud



Decentralized Autonomous Organization

- Key Features

- **Decentralization**

- No central authority → Power and decision-making are distributed among community members, typically those who hold the DAO's native tokens.

- **Autonomy**

- DAOs function through smart contracts (self-executing code on the blockchain) that automate rules, proposals, and transactions once certain conditions are met.

- **Transparency**

- All actions, votes, and transactions within a DAO are recorded on the blockchain, making them publicly auditable and tamper-resistant.

- **Token-Based Governance**

- Members gain voting rights by holding the DAO's tokens. The more tokens a member holds, the greater their voting power, though some DAOs implement mechanisms to prevent dominance by large holders.

- **Open Participation**

- Anyone can join a DAO by acquiring its tokens, participate in governance, and propose or vote on initiatives



Decentralized Autonomous Organization

- How a DAO Works
 - **Smart Contracts**
 - The foundational rules and mechanisms of the DAO are written into smart contracts and deployed on a blockchain.
 - **Funding and Token Distribution**
 - DAOs often raise capital by issuing tokens, which grant holders participation and voting rights.
 - **Proposals and Voting**
 - Members submit proposals for changes or actions. Token holders vote, and if a proposal meets the required threshold, the smart contract automatically executes the decision.
 - **Execution**
 - Actions such as fund allocation, code updates, or project launches are carried out automatically by the smart contracts, without human intervention



Decentralized Autonomous Organization

- Use Cases
 - *Cryptocurrency governance (e.g., MakerDAO for stablecoin management)*
 - *Collective investment funds*
 - *NFT and digital asset acquisition*
 - *Decentralized social platforms*
 - *Virtual worlds and gaming economies*

Decentralized Autonomous Organization

- Ups & Downs
 - + *Increased transparency and trust*
 - + *Reduced need for intermediaries*
 - + *Automated, efficient operations*
 - + *Open, global participation*
 - *Security risks (bugs in smart contracts can be exploited)*
 - *Legal and regulatory uncertainty*
 - *Potential for governance manipulation by large token holders*



Smart Contract

- a computer protocol intended to
 - digitally facilitate, verify or enforce the negotiation or performance of a contract



Smart Contract

- *a new class of apps*
- *supports agreements between parties*
 - *when conditions agreed are satisfied contract is automatically executed*
- *No proxies / no middleman / no lawyers*
- *linked together creates DAOs*
- *DAOs – are companies that do things automatically*

Blockchain applications – in a nutshell

Technology that enables exchange of:

- Ownership titles
- Medical records
- Cryptocurrencies



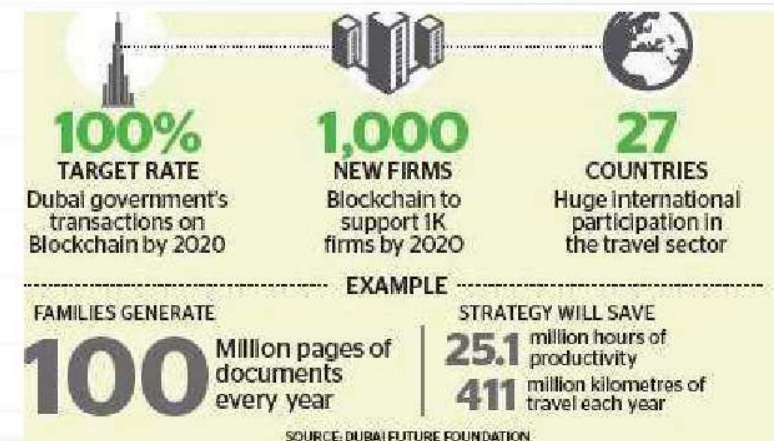
Blockchain applications - Entertainment

- *View.ly (a competitor to YT)*
- *Viuly.com / Flixxo.com (viewers get paid to watch)*
- *d.tube*
- everipedia.org
 - Co-creator Larry Sanger (Wikipedia founder)
- EOS.IO

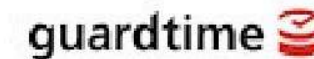


Blockchain applications - Government

- **Dubai** – world's first blockchain powered state
 - Est. 1,5 billion dollar of savings per year stem from moving to a paperless
- **Estonia** (partnered with Ericsson) – all public records stored in BC
- **South Korea** (Samsung) – BC use in public safety and transport apps
- **UK** – BC will be used for pension payments



Blockchain applications – Cybersecurity



- **Guardtime** – *keyless signature systems*
 - a digital signature system based on blockchain technology
 - secures medical records (to date 1 million Estonian patients)



Blockchain applications – Healthcare

- ***GEM (Ericsson) – disease outbreak data***
 - To increase effectiveness of disaster relief and respond
- ***Simply Vital***
 - to track to patients' state after they leave the hospital
 - decentralized patients' records
- ***MedRec (MIT)***
 - Authentication, confidentiality, data sharing of medical records (Israel)



Gem



Blockchain applications – Financial Services

- **BARCLAYS BANK**

- tracking financial transactions
- combating fraud
- shorten the time of transaction verification to <4 hrs from 7-10 days



- **BitPesa (Kenya)**

- a digital foreign exchange and payment platform for frontier market
- allows reduce* cost to send money donated to charity

*according to U.S. Bureau of Economic Analysis (BEA) in 2017 total cost to send money to sub-Saharan Africa was over 30 billion US dollars (!!!)



Blockchain applications – Manufacturing and Industrial

- ***JIO***

- *a mobile operator*
- *owned by Reliance Industries - India's conglomerate of companies*
- *developing a BC based supply chain logistic platform*
- *own cryptocurrency – JIO Coin*



- ***TransActiv Grid (NY)***

- *Allows users to produce and sell energy*
- *Reduces cost of energy (cut on distribution costs)*



Blockchain applications – Manufacturing and Industrial

- ***STORJ.IO***

- Distributed and encrypted cloud storage
- Allows members to share and use HDD space



Blockchain applications – Retail / Real estate / Transport

- ***OPEN BAZAAR***

- Decentralized market for services and goods
- No market fees



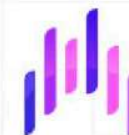
- ***Ubitquity.io***

- BC tracking system for tracking complete legal process in real estate market



- ***Arcade City***

- Uber like, ride-sharing BC based app



Ethereum vs. BTC

- A blockchain implementation
- **Ether** is a cryptocurrency
- Believed to be a dominant platform in long term
- **Advantages**
 - Blocks created faster (~12-14' vs. 10")
 - Other applications aside of currency
 - Easier to mine (encourage individual miners over pools)
 - Uses Turing complete programming language

 VS 		
Founder	Satoshi Nakamoto	Vitalik Buterin
Release Date	9 Jan 2008	30 July 2015
Release Method	Genesis Block Mined	Presale
Blockchain	Proof of work	Proof of work (Planning for POS)
Useage	Digital Currency	Smart Contracts Digital Currency
Cryptocurrency Used	Bitcoin(Satoshi)	Ether
Algorithm	SHA-256	Ethash
Blocks Time	10 Mintues	12-14 Seconds
Mining	ASIC miners	GPUs
Scalable	Not now	Yes

Ethereum vs. BTC

- **Disadvantages**

- Smaller market / value
- Small awareness
- Ghost Protocol (helps eliminates pool mining)
- Unlimited supply of coins



Ethereum Classic

- *Infamous Hack** – DAO Hack (50 mln USD lost)
 - hack undone by hard fork
 - certain members opposed it
 - against concept of cryptocurrencies – never changeable ledger
 - most users / developed moved to new chain Ethereum - old one (Classic)



* <https://www.coindesk.com/understanding-dao-hack-journalists>

Ripple (XRP)



- Ripple Network **!=** Blockchain (a payment settlement system and currency exchange)
- money transfer network designed to serve the needs of the financial services industry
- No mining – pre-mined 100 billion tokens
- Consensus (not user faced) network
- Used by big institutions like banks
 - Intermediary for asset transfer around the world
- Debt based system
 - like conventional banking system

Ripple (XRP)

- Not completely distributed
 - Most of validation made by Ripple Labs
- Like Visa / PayPal with BTC philosophy behind
- Public P2P transactions
- No central authorities



Ripple (XRP)



- **Advantages**

- **Fast settlement.** Transaction are fast. 4-5 seconds, compared with the days it may take banks to complete a wire transfer or the minutes or potentially hours it takes for Bitcoin transactions to be verified.
- **Very low fees.** The cost to complete a transaction on the Ripple network is just 0.00001 XRP, a small fraction of a penny at current rates.
- **Versatile exchange network.** The Ripple network not only processes transactions using XRP. But it can also be used for other fiat currencies and cryptocurrencies.
- **Used by large financial institutions.** Large enterprises can also use Ripple as a transaction platform. Santander and Bank of America are a few using this network, demonstrating it already has larger institutional market adoption than most cryptocurrencies.



Ripple (XRP)



- **Disadvantages**

- **Somewhat centralized.** One of the reasons that cryptocurrencies became popular is that they were decentralized, taking control away from large banks and governments. The Ripple system can be somewhat centralized because of its default list of validators, which goes against this philosophy.
- **Large pre-mined XRP supply.** Though most of the Ripple supply not held in circulation is stored in escrow, it's possible large quantities may get introduced at inopportune times, which could impact XRP's value.
- **SEC action against XRP.** In December 2020, the SEC filed a lawsuit against Ripple, saying that since it can decide when to release XRP, the company should have registered it as a security. The company has denied the allegation.



LiteCoin (LTC)

- **Advantages**

- based on BTC Core
- lightning network / SegWit
- greater number of transactions to process in a second (faster x4)

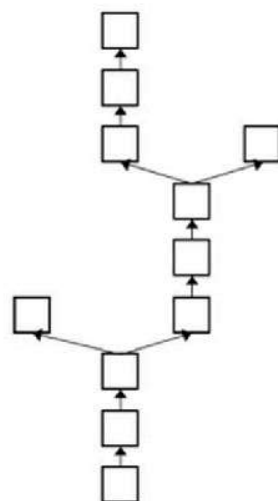
- **Disadvantages**

- uses “*script*” in its **PoW** algorithm, a sequential memory-hard function requiring asymptotically more memory than an algorithm which is not ‘memory-hard’
- FPGA and ASIC devices made for mining are more complicated and more expensive to produce

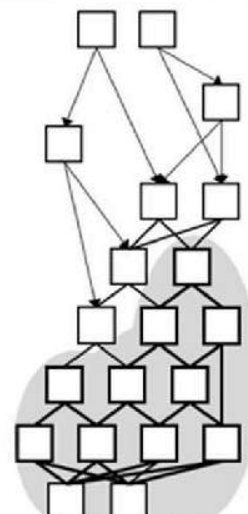
Tangle



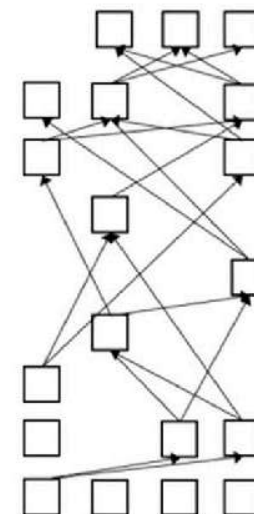
Blockchain



Tangle



Hashgraph



Technology

Block chain

Directed acyclic graph

Directed acyclic graph

Copyright

Open source

Open source

Patented

Consensus

Proof of Work: SHA256-Hash

Proof of Work: check of Tangle tip

Virtual voting

Openness

Public ledger

Public ledger

Private ledger

Applications

Bitcoin

Iota

Swirlds

Efficiency (tps)

3-4

500-800

> 250,000